

1) Tell me about yourself?

Ans ⇒ My name is Kapildeo Patil, Basically I am from Chandrapur, Maharashtra. I completed my graduation from Savitribai Phule Pune University in 2017.

Now I am working in SVN as a Linux admin for 1 year, previously I worked in Credit Vidya as a Linux support for 3.5 years. I have technical experience on Linux, AWS cloud, Nagios, Microsoft teams, slack, Jira, Putty, Git and Basic Shell Scripting.

the alerts related to infrastructure resource, network connectivity, firewall, performance, package, user, permission, authentication, password and LVM management. I am responsible for Documentation, escalation and coordinating with different teams to resolve issues efficiently.

We maintain the SLA report & incident report by using the Jira ticketing tools as well as maintain the google sheet for Shift handover. In the SLA report we maintain records of the production uptime/downtime for daily basis. I was sending SLA report to the manager & vendor. This is all about my work experience, skills and education.

2) Have you worked on Linux? What is Linux basic commands?

Ans ⇒ traceroute, mtr, ping, dmesg, systemctl, yum, ssh, top, ps, awk, sed, cut, lvm, fdisk, rm, rmdir, mkdir, diff, chmod, chown, cd, ls, touch, vi, vim, cat, find, grep, du, df, free, lsblk, etc.

Nmap: - Network exploration tool and security scanner for discovering hosts and services.
Scan all ports: nmap -p 22 192.168.1.1

tcpdump:- Captures and analyses network packets. Capture packets on an interface: - tcpdump -i eth0
Capture packets with a specific port: - tcpdump -i eth0 80

route: - Shows information about the route table it consists of Destination ip address, Gateway, Genmask, User and Interface. Display the route table: route -n
Add a new route: - sudo route add -net 192.168.2.0/24 gw 192.168.1.1

curl: - download & send data from a server using various protocols. wget:- download data from a server using various protocols.

grep: - command in Linux is used for searching text using patterns. It's a powerful tool for filtering and finding specific strings or patterns in files or output.

traceroute - The traceroute command is used to trace the path that packets take from your machine to a destination host. It helps to identify the route and delays at each hop (journey) along the network path. The * * * indicates a hop where the packet was blocked or the router didn't respond.

mtr -The mtr command is a combination of ping and traceroute commands. It is a network diagnostic tool that continuously sends packets showing ping time for each hop is like ping. It also displays network problems of the entire route taken by the network packets like traceroute. It provides real-time analysis of the path packets take to a destination, showing the latency and packet loss at each hop along the route.

-r: set number of cycles and then display the results. mtr -r jupiter.com

-i set the interval between sending each packet. (default 1 sec)

-s: The default is 64 bytes. Set the custom packet size.

-u: Sends UDP packets instead of ICMP Echo requests. This is useful for tracing paths in networks that block ICMP traffic.

-4: This is useful in environments where you want to test IPv4 connectivity.

-6: This is useful in environments where you want to test IPv6 connectivity.

-a: Specific source IP address used to send the packets. mtr -a 192.168.1.100 jupiter.com

-b: Binds to a specific source port. This is useful when testing connectivity from a specific port.

ping: Checks network connectivity by sending ICMP Echo Request packets. The ping command is a

network utility used to test the reachability of a host on a network. It operates by sending Internet Control Message Protocol (ICMP) echo request packets to the target host and waits for an echo reply. The command measures the time it takes for the packets to travel to the host and back, providing valuable information about the network's status and latency.

3) What is a file system and how to Increase & decrease the Size?

Ans ⇒ A file system is the method and structure an operating system uses to store, organize, manage and retrieve data on storage devices like hard drives and SSDs.

Functions of a File System:

Store data - Saves and organizes files and directories.

Track usage - Keeps metadata like size, ownership, permissions, and timestamps. Manage space - Tracks used and free space on the storage device.

Access control - Enforces permissions and user access. Structure - Provides a hierarchy (folders and subfolders).

	Ext2	Ext3	Ext4	XFS
Max File Size	2-4 TB	2 TB	16 TB	8 EB
Max Storage Size	32 TB	16 TB	1 EB	8 EB
Performance	slow	fast>ext2	fast>ext3	fast>ext4

XFS: Excellent for large files, databases, or high I/O workloads.

Note: - When %util is close to 100% in iostat, but IOPS are low, the disk is likely underperforming or overloaded.

Step 1- Ext1,2,3,4 or xfs file system ⇒ `sudo growpart /dev/xvda`

Step 2- Ext1,2,3,4 file system ⇒ `sudo resize2fs /dev/xvda`

Step 2- xfs file system ⇒ `sudo xfs_growfs -d /dev/xvda`

It erases all existing data on the partition. After formatting, the partition is empty but now structured to store files using the ext/xfs format.

4) What ps stand for?

Ans ⇒ The ps command in Linux stands for "Process Status". It is used to display information about active processes running on the system.

5) What is patching? And how will you perform?

Ans ⇒ Before the update/patching process we take an AMI/VMware snapshot. In case of failure to support the updated version, we can easily restore. Patching is the process of updating software, operating systems, or applications to fix bugs, security vulnerabilities, or improve performance. In Linux, patching typically involves updating installed packages, applying kernel updates, or manually applying software patches.

6) Do you have knowledge of AWS?

Ans ⇒ yes, EC2, SG, volume, IAM user Management, IAM policies, MFA, CloudWatch, CloudTrail, SNS, load balancer, auto scaling etc.

7) Explain the ulimit command?

Ans ⇒ The ulimit command in Linux is used to control the system resource limits for user processes. It allows restrictions on CPU time utilization, file sizes, memory usage, open files and more. It helps prevent a single user or process from consuming too many system resources, which can cause system slowdowns or crashes.

8) How to make passwords less connections or authentication?

Ans ⇒ Password less connections in Linux can be achieved using SSH key authentication. This allows secure remote access without entering a password each time. We must disable password-based authentication and enable key based authentication in `/etc/ssh/sshd.conf`.

9) Soft link and hard link?

Ans ⇒ Hard Links: - A hard link is essentially a direct reference to the data on the disk. When you create a hard link to a file, it creates another directory entry for that file, but both the original and the hard link point to the same inode. If you delete the original file, the data is not deleted until all hard links to the file are deleted. Since hard links share the same inode, they do not occupy additional disk space. You can see this using `ls -li` Create hard link = `ln -i target-file file2`

Soft Links: A soft link is a special type of file that acts as a pointer or reference to the main file or directory. soft links point to the file name rather than the inode. If the original file is deleted, the soft link becomes "broken" and does not point to valid data.

Different Inode Numbers: A soft link has its own inode, different from the original file. -s it Create symbolic link = `ln -s target-file file2`

10) What is the inode?

Ans ⇒ inode (index node) is a unique number of file or directory, it is a data structure used by the filesystem to store information about a file or directory, except its name or actual data.

File type (regular file, directory, symbolic link, etc.) Permissions (read/write/execute for owner, group, others) Owner (UID), Group (GID), File size

Timestamps: change time, modification time, last access time.

Link count (number of hard links to the file)

Pointers to data blocks (addresses where the actual file data is stored on disk)

Every file has a unique inode number within its filesystem. When you use `ls -li`, it displays the inode number.

The number of inode is created at the time of formatting the disk (`mkfs.Ext3`). It is dependent on the file system Ext2/Ext3 default size 128 KB, Ext4 default size 256 KB & xfs default size 512 or above KB.

total inode No. = storage / size

11) Reason for job change?

Ans ⇒ I have learned a lot here and it has been a great time but to grow my career, I need to expose myself with the new challenges. So, I am looking for new opportunities to expand my skills and knowledge. I am ready to take more responsibility but unfortunately, there are no leadership opportunities in my current organization. I feel this position is a better match with my skills, knowledge & experience. I am looking forward to bringing my skills, knowledge & experience through my previous job to your company.

12) you have hit rm command on server and deleted everything, now what will you inform client and your manager and how?

Ans ⇒ 1. I will Stop all activity on the server to prevent further damage or overwriting of deleted data. 2. Secondary i will Check if backups exist that can be restored.

To Client: I will be transparent and professional with the Client. Inform them about the incident and its impact. I assure them that our team is actively working on it for the recovery. Provide an estimated timeline for resolution and provide updates from time to time.

To Manager: Provide a factual report, including what happened, why, and what steps are being taken by me. I will take responsibility but also highlight recovery actions. Suggest preventive measures to avoid such incidents in the future.

13) What is DHCP and dns server?

Ans ⇒ Dynamic Host Configuration Protocol (DHCP): - 67 Used by the server & 68 Used by a client. It is a network management protocol that automatically assigns IP addresses and other communication parameters to devices on a network such as DNS addresses, subnet masks, and default gateways. DHCP uses a client-server architecture to assign addresses to devices, such as desktop computers, laptops, tablets, and mobile devices.

New IP addresses: DHCP automatically assigns new IP addresses when devices move to new locations on the network.

Drawbacks:

Security: - DHCP servers can't authenticate clients that request IP addresses, so clients can gain access to unauthorized IP addresses.

No static IP - Computers connected to a network with DHCP can't be used as servers because their IP

address is always changing.

DHCP Server: A server that manages the pool of IP addresses and configuration information. It assigns IP addresses to clients and tracks which IPs are in use and which one not in use.

DHCP Client: A device (computer) that requests and receives configuration information from the DHCP server.

DHCP Lease: The time duration for which an IP address is assigned to a client. Once the lease expires, the client must renew it or request a new IP.

Automatic IP Assign: Reduce the manual configuration, minimizing human error. Centralized Management: Allows network administrators to manage IP address allocation & related configuration from a single point.

Efficient Use of IP: Ensures that IP addresses are reused and managed efficiently, preventing IP conflicts and conserving address space.

Static IP: An IP address is manually assigned to a device and remains constant. This is useful for servers, printers and other devices that need a fixed IP for consistent accessibility.

DNS: is a protocol that is used to translate human readable domain names (google.com) into ip addresses (192.0.1.90) that computers identify each other on the network. It is referred to as the "phonebook" of the internet.

A record: - Maps a domain name to an IPv4 address.

AAAA record: - Maps a domain name to an IPv6 address.

CNAME record: - Redirects one domain name to another domain name (alias).

14) how to configure ssh login? How to change the ssh port number?

Ans ⇒ add ssh port in inbound rule(sg). Create a pem or ppk file as per your requirement to authentication while accessing the remote server. /etc/ssh/sshd.conf and uncomment line and mention your port number.

15) What is BIOS? Booting process? Explain Run levels in Linux?

Ans ⇒ BIOS ⇒ bios stand for basic input output system; bios execute the post operation. Post stands for power on self-tests. In this stage it will check all hardware components like the cpu, memory, storage devices, input and output ports connection is working or not.

MBR ⇒ MBR stands for master boot record, it is 512 bytes of memory divided into three-part boot loader, partition table and error detector. Out of 512-byte 446 byte occupied by the boot loader, 64 bytes occupied by partition table and 2 bytes occupied by error detector.

GRUB ⇒ grub stands for grand unified boot loader, in this stage boot loader is used to load the Linux kernel into the memory and give control to the kernel.

Kernel ⇒ kernel is the heart of the operating system; it is the interface between the hardware and software. It provides necessary service to hardware and software. It is like a bridge between software and hardware.

Init ⇒ init stands for initialization, in this stage load & activates all applications, hardware, software and network devices and then it gives control to run level.

Run Level ⇒ there are 7 run levels.

- Shut down machine
- Single user mode (used for the recovery/Maintenance/Rescue mode)
- Multiuser without NFS
- Mostly used for server
- Not in used (Custom use, rarely used)
- Multiuser with GUI (used for desktop)
- Reboot OS.

Login Screen & User Session ⇒ Once all system services are started, the login prompt (CLI or GUI) appears, and the user can access the system.

16) how to transfer files between two servers without using ftp?

Ans ⇒ scp (Secure Copy Protocol): - scp is a simple and secure way to transfer files over SSH.
rsync (Remote Sync): - rsync is more efficient than scp because it transfers only the differences between source and destination.

Use ssh with tar for Large File Transfer: - combine tar with ssh to compress and transfer files.

17) How to create an archive file?

Ans ⇒ If you're on Linux, you can use tar or zip to create an archive.

tar -cvzf archive_name.tar.gz file or folder

-c: Create an archive, -z: Compress with gzip

18) how to check connectivity of one server to another server?

Ans ⇒ Using the ping command we can check Basic Connectivity. Send ICMP echo requests to the target server. If replies are received, the server is reachable. If the request times out, the server may be down or ICMP might be blocked, internet issue, connect with VPN before trying to access the dashboard.

19) What are the daily activities?

Ans ⇒ We monitor infra resources, monitor changes in the inbound and outbound rules, IAM user activity, systems and networks, CPU usage, memory utilization, connectivity (ping), apache/nginx web (httpd) service, network traffic and firewall. After getting this alert I troubleshoot and find RCA and preventive solution and for unresolved issue I Escalate Metrics to next level, I sent it to the teams channel and connect on bridge call if necessary.

we maintain the google sheet for Daily Shift handover. In the SLA report we maintain the production uptime and downtime for daily and at the start of every month I send an SLA report to the manager including senior manager & vendor.

Shift Handover: At the end of the shift, we provide detailed handover reports to the next teammate, summarizing the activities, issues, and actions taken during my shift. Maintain accurate and up-to-date documentation for all systems, processes and incidents.

20) How will you manage the incident?

Ans⇒ when we get any critical issue I send it to the team's channel and try to resolve that issue and the issue that is not resolved We escalate that Metrics to a L-2 level. if it is not resolved within 10 minutes. In that case I will take the next steps and reach the senior level team, and I will follow up the incident until it is resolved. Using the Jira ticketing tool, we maintain records about incidents, actions taken, resolutions, RCA and preventive solutions. Generate and review daily reports on system performance, incidents and any other relevant metrics.

Generally, I troubleshoot alerts related to Memory, CPU, load, total process, server add & remove, Email add & remove, restart, stop & start Service when it is necessary.

21) port number of ftp, sftp?

Ans ⇒ The default SFTP (Secure File Transfer Protocol) port is 22, the same as SSH. SFTP runs over SSH, so check the port in the SSH configuration file. If no custom port is set, SFTP runs on port 22. If a custom port is set "2222" use that instead of default.

FTP: Port 21 is used for the control connection in FTP, where commands and responses are exchanged, but it does not transfer or receive files directly. Port 21: Handles the communication between the client and server for sending commands (Ex. ls, get, put) and receiving responses (like directory listings). Port 20: Handles the actual transfer or receipt of data, such as files being uploaded or downloaded.

22) How do you configure Nagios monitoring tools? How to set up thresholds and alerts?

Ans ⇒ check git Nagios.sh file. Add host name, service name, ip address, check command (threshold)
Define a host to the local machine

```

define host {
use linux-server host_name jenkins-server alias nagios-nrpe
address 3.87.172.140
}
# Define a service to the local machine
define service {
use generic service
host_name jenkins-server service_description Check Apache check_command check_http
}

```

23) Do you know the use of the commands awk, Sed and cut?

Ans ⇒ Awk (Pattern Scanning & text Processing) ⇒ Used for processing structured text (like logs, CSV files, etc.). Works on columns and fields in text files. Can perform calculations, filtering, and text transformations.

Sed (text editor) ⇒ Used for modifying text in a file. Works line-by-line and supports find & replace, deletion and insertion.

Cut (Column Extraction) ⇒ Extracts specific columns or fields from a file. Works faster than awk but it is less flexible.

24) Where do the logs generate? How to check hardware logs?

Ans ⇒ /var/log/auth - system - user famous command to check hardware logs dmesg

25) What is fstab and what is the use of fstab?

Ans ⇒ fstab (File System Table) is a configuration file in Linux that contains information about disk partitions and other file systems. It will mount automatically when the system starts up. It is located at: /etc/fstab

The fstab file defines how storage devices and partitions should be mounted and managed. It ensures that specific filesystems are mounted in a consistent manner every time the system boots.

Filesystem - The partition or device to be mounted (/dev/sda1, UUID=xxxx).

Mount Point - The directory where the filesystem will be mounted (/mnt/data, /home, /).

Filesystem Type - The type of filesystem (ext1,2,3,4, xfs).

Mount Option - it defines how the filesystem should be mounted (defaults,ro/rw, noexec).

Dump - Used by the dump utility for backups (0 = ignore).

check filesystems at boot- check filesystem (0= disable, 1= enable)

26) How to check disk usage?

Ans ⇒ du -hs /home/* ⇒ show all directory consumes memory. du -hs /var/www/html ⇒ show the specific folder memory.

df -hT ⇒ show all hard drives consume memory, available, file format, mount location etc. (df -ht /dev/xvda1 ⇒ for particular disc).

free ⇒ display physical memory (RAM) and swap space. lsblk ⇒ display the hard disk and partition.

27) How to check the running process and CPU utilization?

Ans ⇒ top and ps is a famous tool for checking these metrics.

28) Explain about APM?

Ans ⇒ APM (Application Performance Monitoring) is a set of tools and practices used to track the performance, availability, and user experience of applications in real time. It helps in identifying and diagnosing performance issues, ensuring optimal application performance, and improving overall user satisfaction.

APM tools collect key metrics:

Response time - How long an application takes to respond to a request.

Throughput - The number of requests processed per second.

Error rate, Error logs - The percentage of failed requests.

CPU and memory usage - The system resources consumed by the application.

Disk I/O and Network latency - Measures input/output operations and network delays.

Alerting and Notifications

Sends alerts when performance thresholds are exceeded. Can trigger automated responses for issue resolution.

Notifies teams via email.

Popular APM Tools

Nagios - infrastructure, application and performance monitoring.

Prometheus & Grafana - Open-source monitoring with powerful visualization.

29) Which type of SLA policy do you follow?

Ans ⇒ low severity issue resolution within 20 minutes. medium severity issue resolution within 15 minutes. Critical severity issue resolution within 10 minutes.

30) Which types of metrics do you observe in CloudWatch?

Ans ⇒ EC2 Metrics

CPU Utilization → Percentage of CPU usage DiskRead / DiskWrite → Number of disk I/O operations

Network In / Network Out → Data transferred in and out StatusCheckFailed → Instance status health check S3 Metrics

Bucket Size Bytes → Total bucket size Number of Objects → Total objects in a bucket

Get Requests / Put Requests → Count of GET and PUT requests

Alert consists of account, metric name, datapoint(threshold), state changes, time millisecond.

31) How do I check the number of CPU Cores in my system?

Ans ⇒ Using cat /proc/cpuinfo

Cat /proc/cpuinfo | grep -c "processor"

32) What are Infra Metric servers?

Ans ⇒ Infra Metric servers refer to servers that collect, monitor, and analyse infrastructure-related performance metrics, including CPU usage, memory consumption, disk usage, network activity, and application performance. These servers help IT teams manage system health, detect issues early, and optimize resource usage.

33) What troubleshooting issue you faced and what was the root cause? Any Critical Incident ?

Ans ⇒ server was not accessing Cause SSL Certificate Expiration.

One day, due to an SSL certificate expiration, employees were unable to access the server. This was a major issue as it disrupted normal operations. We identified the problem and renewed the SSL certificate on the GoDaddy website. During this process, we experienced 2 minutes of downtime before the issue was fully resolved.

Scenario: - A major client reported that users were unable to access their web application, receiving SSL/TLS security warnings (e.g. "Your connection is not private") when visiting the site. At the same time, our monitoring system detected multiple SSL handshake failures in the logs. After resolving the issue, we documented a Root Cause Analysis (RCA) in our ticketing system and implemented proactive SSL monitoring alerts to prevent future occurrences.

34) What is the command to create a partition and check the partition?

Ans ⇒ Create a MBR based Partition Using fdisk: - sudo fdisk /dev/xvda follow these steps:

Press n (New partition).

Choose p (Primary) or e (Extended). Select a partition number.

Specify the start and end sectors (or press Enter to use defaults).

Press W to write changes and exit.

By using lsblk & fdisk -l, for mounted partition by using df -h

sudo fdisk /dev/xvdf ⇒ create partitions

sudo partprobe /dev/xvdf ⇒ inform OS about the changes happened in partition

Create a GPT based Partition Using gdisk:- sudo gdisk /dev/xvda

35) How do you configure the partition is mandatory and which is present at the time of booting?

Ans ⇒ To ensure that a particular partition is mandatory at the time of booting, we can configure the partition in the /etc/fstab file. The /etc/fstab file is responsible for defining which partitions should be

mounted at boot and how it should be mounted. Add a new entry for the partition at the end of the file. Use either the device name or UUID to ensure it is properly identified.

36) Apache & nginx configuration?

Ans ⇒ I will install the package by using the yum command. Then I Enable and start the apache service using systemctl command. Then we check the service status. Then the firewall will allow 80/443 ports in the inbound rule for apache/nginx and 8080 ports for tomcat. Apache and Nginx both are open-source web servers that are commonly used to host websites and applications.

Nginx apache

Performance Excellent Slower under heavy load

Resource Usage Low Higher

OS Compatibility Best on Unix/Linux, limited on Windows Cross-platform Reverse Proxy Built-in, highly efficient Supported but slower

37) What is the difference between rpm and yum?

Ans ⇒ RPM Manages individual package files. Does not resolve dependencies automatically.

Whenever we have to install any package, we have to run commands into the particular folder where the package is located.

Yum installs packages and automatically resolves dependencies. Install packages from any location.

Downloads packages from configured repositories. Use online repositories for package installation and updates. Easier, as it handles dependencies automatically.

Use Case

Use rpm when you must install and remove a specific package manually.

Use yum when you want to install or update packages while automatically resolving dependencies.

38) How will you document all issues and how will you follow up the handover?

Ans ⇒ Documenting issues systematically is crucial for troubleshooting, tracking, and resolving problems efficiently. Proper documentation and handover ensure that issues are tracked, resolved, and communicated effectively across teams. We use a Jira software tool and google sheet for maintaining information about incident metrics, resolution action, RCA, priority, open and close issue status.

Pending Issues: - List open issues with status, priority, and owner. Ensure the next team understands pending tasks.

Resolved Issues & Fixes: - What was fixed? How? Any remaining risks?

Logs & Monitoring Details: - Error logs, system health reports, API monitoring dashboards. Next Steps

& Follow-ups: - What actions should the next team take? Or any scheduled maintenance or deadlines?

Shift Handover: At the end of the shift, we provide detailed handover reports to the upcoming shift, summarizing the activities, issues, and actions taken during my shift. Maintain accurate and up-to-date documentation for all systems, processes and incidents.

39) What are your roles and responsibilities?

Ans ⇒ I am responsible for system performance, server stability, security, server availability, incident management, unresolved incident metrics Escalation, respond to alerts and incidents, Documentation & Reporting, maintaining and troubleshooting network and IT infrastructure to ensure smooth and uninterrupted operations. The role involves real-time network monitoring and issue resolution. Server outages, latency, failures, System Health Checks troubleshoot CPU, RAM, disk space, and network traffic issue.

Security Monitoring - SG, vpn, user activity and unauthorized access. Performance Optimization - Analyse logs and suggest improvements.

Documentation - Maintain logs, create incident reports, RCA and improve response plans.

40) Difference b/w Unix and Linux?

Ans ⇒ unix:- Developed in the 1960s-70s at AT&T Bell Labs by Ken Thompson and Dennis Ritchie. Primarily command-line interface, but GUI options exist. The original Unix source code is not freely available.

Linux: - Created by Linus Torvalds in 1991 as a free and open-source alternative to Unix.

Open-source - The Linux kernel is freely available and modifiable. Offers both CLI and GUI.

41) Difference between EBS volume and AMI?

Ans ⇒ EBS Volume: - It is a block storage device that provides raw storage for EC2 instances. Used as a virtual hard drive for EC2 instances to store data persistently. Data remains even if the EC2 instance is stopped or terminated (unless it is the root volume and delete on termination is enabled for root volume). Attached to an EC2 instance as primary (root) or secondary storage.

AMI (Amazon Machine Image): - A pre-configured image containing the OS, application, and configuration used to launch EC2 instances. Used to create and launch new EC2 instances with pre-installed software and configurations. AMI itself is persistent and can be used repeatedly to launch instances. Used to deploy multiple instances with the same configuration.

42) What is Infrastructure?

Ans ⇒ Infrastructure, refers to the foundational hardware, software, network, and services that support IT operations in an organization. It includes everything needed to run applications, store data, and enable business processes.

42) What is a hypervisor?

Ans ⇒ hosting the virtual machine. A hypervisor is software that allows you to run multiple virtual machines (VMs) on a single physical machine. Every virtual machine has its own operating system and applications. The hypervisor allocates the physical computing resources such as CPU and memory to individual virtual machines as per requirement.

Additional skills learned or not? Aws, docker, Grafana, Prometheus,

Have you done any certification? Aws

What do you know about the company? Ans ⇒

How do you configure Nagios setup in your infra?

What is Nagios? What is the use? What is the benefit of Nagios monitoring tools?

How to Extend file size 500GB?

Do you know about docker?

Do you work on a network?

If there is no Apache what will u do? Apache service is not starting; how do you fix it?

What are the fields shown in the top command?

Which monitoring tool are you using and what did you see in this monitoring tool?

What is the pre-check of OS Upgradation (RHEL 7 To RHEL 8)?

How do you extend the partition using LVM explained step by step?

How will you increase the physical volume in LVM?

What is LVM? how to extend and reduce LVM partition space?

Shell scripting?

If the server is not booting up, what is the reason? How will you troubleshoot that issue?

The system is rebooting continuously, what may be the problem?

How are you helping to monitor logs in the current organization? How will you help your team to troubleshoot those error logs?

Difference between telnet & scp?

What will you do if the backup fails?

Where are the user details?

Difference between /etc & /passwd file?

What step will you take? If the production server does need to reboot?

If the server performance is low how to identify it as a hardware or software problem?

If the admin account is locked, how will you access the system?

How to lock a user?

How to schedule the cron jobs?

The system is not working fine; how do we approach the client partition extension?

Have you faced any customer related issues?

In the last 6 months what have you learned?

How many ways to create a file? How do you check boot logs?

What is static Ip or how to check static Ip? Back-up in Linux?

How to identify if the router is not working?

43) Find a file content with some word only displaying the file name?

Ans ⇒ using grep command. (grep -i happy kapil.txt)

44) How to enter in single user mode?

Ans ⇒ Reboot the System

Access GRUB Menu by pressing f8 /Esc/: - When the GRUB menu appears, select the Linux kernel entry. Press e to edit the boot parameters.

Modify Kernel Parameters: - Find the line that starts with Linux at the end of the line, replace single. Press ctrl + X to boot

System Enters Single User Mode (You will get a root shell)

Perform Required Maintenance

Reboot After Changes

45) After upgrading the package, the package is not working, what is the reason for that?

Ans ⇒ If a package stops working after an upgrade, there can be several possible reasons.

Dependency Issues: - The new version may require different dependencies, and they might not be installed or are incompatible.

Configuration Changes: - The upgrade may overwrite or remove configuration files.

Service Not Restarted or reloaded: - Some services need a restart after upgrading.

Version Mismatch or not compatible: - The upgrade might not be compatible with your OS or other software.

File Permissions Changed: - The upgrade may have changed ownership or permissions.

46) What is a sticky bit? suid, guid?

Ans ⇒ The sticky bit is a special permission in Linux that applies to directories. It prevents users from deleting or renaming files inside a directory unless they own the file or have special privileges (like roots). Normally, if a directory has written permission for a group or others, any user can delete or modify any file inside it. When the sticky bit is set, only the file's owner, the directory's owner, or root can delete or rename the file. This is commonly used in shared directories like /tmp, where multiple users can create files but shouldn't delete each other's files. Using chmod Command
Add sticky bit on a directory: chmod +t /mydir to remove the sticky bit: chmod -t /mydir Using Numeric Mode

Add sticky bit: chmod 1777 /mydir Remove sticky bit: chmod 0777 /mydir

Scenario: Protecting Files in /tmp; - The /tmp directory is world-writable, meaning any user can create files. Without a sticky bit, users could delete each other's files, causing security issues. With the sticky bit only the file owner can delete their own files, even though everyone can write to /tmp

Verifying Sticky Bit Removal; - After removing the sticky bit, check again If 't' is gone, the sticky bit is removed. most common use for /tmp directory and Shared project folders (to prevent accidental deletions)

suid: - when you want to give permission to run a particular command by any user. chmod u+s /use/sbin/useradd (binary file of useradd command) (4777- numeric value) chmod u-s /use/sbin/useradd (remove suid permission) (0777- numeric value)

guid: - when any user creates a file, but you want to change the group ownership automatically that time it will help.

groupadd tck chgrp tck /xyz

chmod g+s /xyz (when the user will create file, own will same but group owner will tck)

47) What is ansible?

Ans ⇒ Ansible is an open-source IT automation tool used for configuration management, application deployment and infrastructure automation. It simplifies system administration tasks by allowing you to automate repetitive tasks across multiple servers. Uses YAML-based playbooks.

48) What are the metrics you suggest for application monitoring?

Ans ⇒ To effectively monitor an application; you should track four key areas: Performance Metrics (latency, response time, throughput i/o response) Error Metrics (failures, exceptions, HTTP errors)

Infrastructure Metrics (CPU, memory, disk, network)

User Experience Metrics (page load time, API response)

49) What is a snapshot?

Ans ⇒ A snapshot is a point-in-time copy of a system, disk, or virtual machine (VM) that captures its current state. It allows you to restore the system to a previous state in case of failure, corruption, or unwanted changes.

50) If you get a CPU high alert, what steps do you follow?

51) Command for change the user & group permission and ownership?

Ans ⇒ `chown kapil text.txt`

52) How to check and kill the process?

Ans ⇒ `top` command to check, `kill` command to delete (-9)

53) What do you do if the server goes down?

Ans ⇒ first I will check the connectivity using ping and try to access through SSH. Since the server is unreachable over the network, I would check whether the VM or instance is still running from the cloud or virtualization platform. If available, I would use the Serial Console or VM Console to access the server directly, by passing the network. From there, I'd inspect the boot process, check system logs, verify the network configuration and SSH service, and identify whether the issue is caused by a network misconfiguration, a failed service, or an operating system problem. If it's a physical server, I'd use out-of-band management such as iDRAC or iLO for similar access.

54) Server is down; how do you fix it?

Ans ⇒ I will check if the server is down or not. I will Ping the Server: - If there's no response, the server might be offline or have network issues. I will try SSH Access: - If SSH is not working, the server might be unreachable.

Check Web Service - If there's no response, the web service might be down. Check Network Issues: - Ensure the correct interface has an IP address.

Check System Logs: - network errors

Restart the Server: - If the issue persists, restart the server

Check Using Port: - Ensure required services (e.g. SSH, web server) are listening. Check Firewall Rules: - If the firewall is blocking connections, disable it temporarily:

Check Server Status: - If you have physical access, check if the server is powered on. If it's a Cloud server, try rebooting from the provider's control panel.

Check System Logs, Check Running Services, Check Disk Space, Check Network Issues.

55) What one thing is the person super proud of in his current work?

Ans ⇒ when he contributed to the company's growth.

56) What is NetFlow?

Ans ⇒ NetFlow is a network protocol developed by Cisco that collects IP traffic information as it flows through a network device, typically a router or switch. It provides valuable data that can be used to monitor, analyse, and optimize network traffic. NetFlow helps to understand network behaviour, improve security, and manage network performance.

57) What is the Zombie process?

Ans ⇒ zombie process is a process that has completed execution but still has an entry in the process table because its parent process has not yet read its exit status. Zombie processes do not use system resources, but they occupy process table slots, which can become problematic if too many accumulate. You cannot kill a zombie process directly because it is already dead. However, you can try these methods by Killing the Parent Process.

58) Is the zombie process good or bad?

Ans ⇒ If too many zombies accumulate, they waste process table slots and can eventually stop to create new processes. A stuck zombie may indicate a bug in the parent process. On critical servers, excessive zombie processes can lead to performance issues.

59) How do you give SUDO permissions to normal users?

Ans ⇒ need to add the user in the /etc/sudoers file.

60) Tell me about the Nginx web server and what are the benefits?

Ans ⇒ Web Server - Serves static & dynamic content efficiently.

Reverse Proxy - Can sit in front of application servers to improve performance. Load Balancer - Distributes traffic across multiple backend servers.

SSL/TLS Termination - Handles secure HTTPS connections.

61) If the server is rebooted suddenly what was the reason? How will you verify that?

Ans ⇒ If a server reboots suddenly, the cause could be due to various factors, including hardware issues, system crashes, kernel panics, power failures, or scheduled reboots.

Check the System Logs

Check Last Reboot Time: - Use the last command to check when the system was rebooted

Check Who Initiated the Reboot ⇒ `last -x shutdown reboot | head -10`

Check for Kernel Panics; - A kernel panic may have caused an automatic reboot. Look for messages like "Kernel Panic" in logs.

`cat /var/log/kern.log | grep -i "panic" cat /var/log/messages | grep -i "panic"`

Check for Automatic Reboots (Cron Jobs or System Updates) :- Sometimes, a scheduled cron job or system update might have rebooted the server. Check cron logs: - `cat /var/log/cron.log | grep -i reboot`

Check Hardware & Power Issues: - If the reboot was unexpected, hardware failures might be involved:

Check Hardware Errors ⇒ `dmesg | grep -i "error\|fail\|fault"`

62) What is the Linux Kernel? What is the difference between a shell and a terminal?

Ans ⇒ The Linux Kernel is the core part of the Linux OS that manages hardware resources and provides essential(necessary) services to the system's applications. A terminal is an interface to interact with the system, and the shell is the program that processes commands entered in the terminal.

63) What is the difference between socket and port?

Ans ⇒ A port is a logical endpoint on a network device that represents an application or service, while a socket is a communication endpoint that involves both a port number and an IP address. A socket is part of a port, and a port number is part of a socket.

Port: - A port is a logical connection that allows two endpoints to communicate. Ports are used to differentiate between different types of traffic, such as emails and webpages. Port numbers specify the type of application or process running.

Socket: - A socket is a communication endpoint that enables data transmission between devices. A socket is made up of a port number and an IP address. Sockets are managed by the operating system.

64) What is Dynatrace in APM?

Ans ⇒ Dynatrace is an AI based observability and application performance monitoring (APM) tool used to monitor applications, infrastructure, networks, and cloud environments in real-time. It provides deep insights into system health, performance issues, and user experience.

65) What is Latency?

Ans ⇒ Latency refers to the time delay between a request being made and the corresponding response being received. It is measured in milliseconds (ms) and is crucial in networking, computing, and real-time systems. Lower latency means faster response times, while higher latency results in noticeable delays.

66) What is the difference between a problem and incident?

Ans ⇒ Incident: - An unplanned event or disruption that affects the normal operation of a service or system. The focus is on restoring the service as quickly as possible.

Example: A website goes down due to high traffic load.

Problem: - The underlying cause of one or more incidents. The focus is on identifying and resolving the

root cause to prevent future occurrences.

Example: The root cause of the website downtime is insufficient server capacity.

67) What is a Bridge Call?

Ans ⇒ A Bridge Call is a conference call where multiple participants can join a single communication session to discuss and resolve issues in real time. It is commonly used in incident management, it involves vendors, IT support, DevOps and NOC teams to troubleshoot critical issues. We use the Microsoft teams for bridge call.

68) How to Diagnose an EC2 Instance in AWS Cloud?

Ans ⇒ When an AWS EC2 instance is not responding, rebooting and performance issues, you can diagnose and troubleshoot using the following steps.

Step 1: Verify Instance State whether it is Running, Stopped or Terminated.

Step 2: Check Instance Reachability (Run SSH command, If SSH fails, check SG and key pair.)

AWS CloudWatch to Monitor EC2 Metrics 1: Check CloudWatch Metrics

CloudWatch → Metrics → EC2

Monitor CPU, memory, disk, and network performance.

2: Enable CloudWatch Logs

Helps in analyzing instance performance over time

Solution for Unresponsive EC2 Instance. 1: Stop & Start the Instance

2: Create a Recovery Instance: - If EC2 is stuck, follow these steps:

Detach the volume from the broken instance. ⇒ Attach it to another working EC2 instance. ⇒

Mount and fix errors. ⇒ Reattach the volume and restart the instance.

69) How to Read Load Average? How much can one CPU handle? How to check load?

Ans ⇒ top, uptime, mpstat -P ALL (CPU load per core)

Load Average: 1.25, 1.05, 0.9

These numbers represent the average system load over 1, 5, and 15 minutes.

Ideal Load for 2-Core CPU: 0.00 - 2.00 → System is running smoothly.

Above 2.00 → CPU is overloaded, and processes are waiting for execution.

70) What is the difference between VPN & Firewall?

Ans ⇒ Firewall: - A firewall is a security system that monitors, filters, and controls incoming and outgoing network traffic based on predefined security rules. Ensures only allowed traffic passes through, based on security rules.

Virtual Private Network: - A VPN creates a secure, encrypted connection between a user device and a remote network over the internet. It is primarily used for privacy, secure data transmission and remote access. Provides secure communication over public networks. Allows users to access private resources remotely. Uses encryption protocols (OpenVPN) to create a secure tunnel between the user's device and the VPN server. Encrypts all traffic passing through the tunnel, ensuring data confidentiality and integrity.

71) What is the difference between SSH & Telnet?

Ans ⇒ SSH: - This is a secure remote login service where data is encrypted. ssh user@ipaddress

Telnet (23): - Connects to remote servers using Telnet protocol. Same function as ssh but data not encrypted. The lack of encryption makes Telnet highly insecure for modern networks. Telnet largely replaced SSH due to its security vulnerabilities.

Connect to a server on a specific port: telnet google.com 80

72) Explain what issue you are troubleshooting?

Ans ⇒ Apache/Nginx - sometimes the application web page is not accessible. That time we restart the Apache service by using the Systemctl command 99 times it gets resolved that issue. CPU & load - we check the CPU % usage and then we check the total running process and i kill those are not important and for those are important i made changes in the niceness for running one by one process and that help to reduce CPU load and smooth operation and it come to normal stage.

Memory - most common issue we face every day, after we receive the alert, we log in to the terminal

and check the memory consumption by each directory and file which is most consumed and later we delete the unnecessary data from that directory and make the free space.

Network and connectivity issue: - In case of a server down we try to check the server connectivity by using ping command after confirmation we restart the server.

Linux patch management: - is the process of updating the Linux operating system and applications to fix bugs, improve performance, and add new features: Patch management is an important part of IT system lifecycle management and vulnerability management.

73) What is RCA and SLA?

Ans ⇒ Root Cause Analysis (RCA) - is a systematic process used to identify the underlying causes of a problem or incident, rather than just addressing its symptoms. The goal of RCA is to determine what caused the issue, so that it can be effectively resolved and prevented from happening again.

Service Level Agreements (SLAs) - SLAs are formal contracts or agreements between a service provider and a customer that define the expected level of service, including specific performance metrics, responsibilities, and any penalties for failing to meet those expectations.

How many alerts do you get daily? Ans⇒ we received 200-300 alerts per day.

Types of Alerts: - Critical Alerts (Requires immediate action) - Server down, network outage, security breach, VPN down, disk storage.

Warning Alerts (Needs investigation) - High CPU usage, increased latency, packet loss, load increased.

Ok Alerts (For Information purposes) - Device reboot, configuration change, log updates. Unknown

Alerts - at the time of adding new services/servers.

Use alert filtering & prioritization - Focus on critical incidents first.

Set up automation-Auto-resolve minor issues with scripts (restart services, move logs). Set up monitoring thresholds - Reduce false alarms from minor fluctuations.

74) How many tickets per day?

Ans ⇒ sir it depends on the alert's priority, average 5 - 10 tickets per day.

75) How do you troubleshoot network alerts?

Ans ⇒ Check the alert details: Review the severity, timestamp and affected devices or services.

Categorize the issue:

Connectivity issues (link down, packet loss) Performance degradation (high latency) Security alerts (unauthorized access)

Hardware failures (interface errors, device overheating) Verify Network Connectivity

Ping the affected device: ping (destination IP)

No response? Possible device down or network failure.

High latency? Possible congestion or routing issue.

Check if a specific port is open (for applications): telnet (destination IP) (port no.)

Use traceroutes to identify path issues: traceroute (destination IP)

76) How do you handle P-1 tickets?

Ans ⇒ P-1 Ticket: - Respond to that alert within 5 minutes. Create the incident ticket by using the Jira software tool. make a call with the vendors & customers if necessary.

Notify stakeholders via email and provide time-to-time updates and inform them about the business impact and estimate time to resolve that issue.

If server-related & security-related involves Devops Engineers. Gather Logs and Troubleshoot Quickly Check affected services, logs, and alerts.

Use monitoring tools (Nagios, Prometheus) to check metrics. Run network checks (ping, traceroute, netstat).

RCA & Resolution: - After resolving the issue, mention root cause and implement permanent fixes. Update documentation for future incident prevention.

77) What is Grafana and how does it work? Have you worked on the Grafana?

Ans ⇒ Grafana is a multi-platform open-source analytics and visualization tool. Support the different

kinds of data sources such as Prometheus, MySQL database, AWS Cloud watch, Google cloud and many more. It can produce charts, graphs, tables and alerts for the web when connected to supported data sources. We can see the server status code, 90 days old data

78) If you get 100 alerts what will do and how will you solve it?

Ans⇒ I will verify it's a false alarm or a real issue. If it is real then I will categorize the alerts based on severity like critical, warning and information (ok) alerts. I will Focus on the high Prioritize Critical Alerts. Use filtering to identify patterns and eliminate noise. Fix monitoring misconfigurations to prevent false alerts.

79) About MIM? Is it full form?

Ans ⇒ Major Incident Management (MIM) is the process of handling critical IT incidents that cause severe (major) business impact, such as network failures, server outages, security breaches & application downtime. The MIM goal is to restore services as quickly as possible with minimum damage. (VPN down, SSL expired).

80) Which monitoring tool are you using or handle?

Ans ⇒ we are using Nagios monitoring tool.

EC2 Instance (CPU Usage, average load, Memory, Disc Space, connectivity)

Secure Sockets Layer (SSL)/Transport Layer Security (TLS)

Monitor SG (inbound/outbound rules change)

Monitor IAM user activity (policy changes, user create or delete, port add or remove, console login fails)

Monitor daily and monthly Cost budget.

Monitor Docker Container and VPN down.

Monitor Application Logs on the Graylag.

81) If you can't solve the problem when alerts are triggered what will you do?

Ans⇒ I will attempt to resolve common issues (restarting web server, clearing disk space, killing unnecessary processes, restart server in case of server down). If the issue persists. I will escalate those metrics to the L-2 team with the detailed Error logs, Error messages, metrics, action taken by me, time duration and impact analysis. Communicate with stakeholders clearly about the business impact, estimate time for the resolution and inform them that our team is working on it continuously via email or team chat. I will follow an escalation and structured troubleshooting approach to ensure a resolution is reached as quickly as possible.

82) Did you handle clients?

Ans⇒ we use the Microsoft teams' tool to initiate bridge calls. through bridge calls we connect with the client, but the senior person only assists the client. We help them in case of need for Error logs, metrics and the time duration (when it occurs).

83) What is the ITIL process?

Ans ⇒ ITIL is a framework for IT service management that helps organizations deliver efficient, reliable, and high-quality IT services. It provides best practices for managing incidents, problems, changes, and service requests in an IT environment.

84) SFTP, Http, Https Protocols, and file share system?

Ans ⇒ HTTP is a protocol used for transferring web pages, files, and data over the internet. It is the foundation of web communication, allowing browsers and servers to exchange information. Supports GET, POST, PUT, DELETE: These are methods for requesting, sending, and modifying data.

Plaintext & Encrypted (HTTPS): HTTP transmits data in plain text, while HTTPS encrypts data for security. By default, both Apache and Nginx typically run on port 80 for standard HTTP connections; for secure HTTPS connections, they will listen on port 443

File Sharing System: - A file-sharing system allows multiple users or devices to access, store and transfer files over a network.

Network File Systems (NFS): - Used for sharing files over a local network. NFS is a distributed file

system protocol that allows users to access files over a network as if they were stored locally. It is widely used in Linux and UNIX environments for file sharing between systems.

Cloud-Based File Sharing: - Services like Google Drive, Dropbox, OneDrive allow file storage and sharing over the internet.

FTP: - Transfers files between a client and a server. Uses FTP/SFTP protocols (using ftp or scp commands in Linux).

85) Difference between tcp and upd protocol?

Ans ⇒ tcp: - connection oriented, reliable, ensure data delivery (http, ftp)

udp: - connectionless, faster, no guarantee of data delivery (DNS)

86) What is TCP & UDP?

Ans ⇒ These are two protocols used for transmitting data over a network. They operate at the transport layer of the OSI model and serve different purposes depending on the requirements of data transmission.

TCP (Transmission Control Protocol)

Connection-Oriented: Establishes a connection between the client and server before data transmission begins. It ensures that both sides are ready for data transfer.

Reliable: Guarantees the delivery of data packets in the correct order. If a packet is lost or corrupted, TCP retransmits it.

Flow Control: Controls the rate of data transmission between the sender and receiver to prevent overwhelming the receiver.

Error Checking: Performs error-checking and ensures data integrity through mechanisms like checksums.

UDP (User Datagram Protocol)

Connectionless: Does not establish a connection before sending data. It sends packets (datagrams) directly to the recipient without checking if they are received.

Unreliable: UDP does not guarantee delivery, order, or error-checking. Packets can be lost, duplicated, or arrive out of order.

No Flow Control: UDP does not manage data flow between sender and receiver, which can lead to congestion if the network is overloaded.

87) Are you okay with a night shift?

Ans ⇒ Yes, I am comfortable working in night shifts. In my current organization, I work in a rotational shift schedule, which includes nights. I am always ready to support my team whenever needed, including nights, weekends, and holidays, ensuring smooth operations and minimal disruptions.

88) What is the ip protocol?

Ans ⇒ The IP protocol stands for Internet Protocol – it's a network-layer protocol that is responsible for delivering data packets from one computer (host) to another across interconnected networks, like the internet.

Addressing: Assigns IP addresses to devices

Packet Routing: Forwards data packets from source to destination
Fragmentation: Breaks large packets into smaller ones to fit MTU

89) What is a private and public network? Type of Ip address?

Ans ⇒ IP (Internet Protocol) addresses are categorized based on their use, function, and version.

Below are the main types. Private network IP addresses cannot be accessed through the internet, but in the public network we can access from the outside.

Based on Version

IPv4 (Internet Protocol Version 4)

Format: 32-bit, written as four octets (192.168.1.1) Limitation: Running out of available addresses.

IPv6 (Internet Protocol Version 6)

Format: 128-bit, written as eight groups of hexadecimal numbers (2001:0db8:85a3:8a2e:0370:7334)

Advantage: More secure and scalable.

Based on Accessibility Public IP Address

Assigned by ISP (Internet Service Provider). Used for global communication (Internet access). Can be static or dynamic.

Private IP Address

Used within local networks (LAN, home, office). Assigned by routers via DHCP.

Not accessible from the Internet. Example Private IP range 192.168.0.0 - 192.168.255.255

10.0.0.0 - 10.255.255.255

172.16.0.0 - 172.31.255.255

3. Based on Assignment

Static IP

Manually assigned by administrator/ISP.

Does not change (assigned permanently).

Used for: Servers, CCTV, remote access.

Dynamic IP

Assigned dynamically by DHCP (ISP or router).

Changes periodically.

Used for: Home networks, most users.

4. Special IP Address Types

Multicast IP Used for sending data to multiple hosts. 224.0.0.0 - 239.255.255.255

Broadcast IP Sends data to all devices in a network. 255.255.255.255

Example IP Classifications

Class A: 10.5.6.7 (Large network)

Class B: 172.16.20.30 (Medium network) Class C: 192.168.1.10 (Small LAN)

Class D: 239.0.0.1 (Multicast streaming)

Class E: 250.1.2.3 (Future use)

90) What are the technical differences between Prometheus & Grafana vs Nagios?

Ans ⇒ Nagios: - Difficult to scale for large environments Monitoring Type is Host & Service-based

Basic alerting with custom scripts

not possible container monitoring (Kubernetes, Docker)

Graphical visualization not possible

Missed some information like, file type, instance id, environment.

Response time is more compared to Grafana & Prometheus.

Prometheus & Grafana: - Highly scalable for cloud-native apps.

Monitoring Type is Metric-based & Time-series

Advanced alerting with Alert Manager

container based monitoring (Kubernetes, Docker).

Graphical visualization possible

Get additional information like file type, instance id, environment.

Fast Response time is compared to Nagios.

91) How to provide hardware support and maintenance for all networking components?

Ans ⇒ Regular Inspections and Testing: Check for physical damages (hardware, cables). Ensure all cables are properly connected and organized.

Performance Testing: Use diagnostic tools to check device performance and latency. Test power supplies and UPS systems to prevent unexpected outages

Software Updates: Regularly update software for all devices to fix bugs and vulnerabilities and enhance performance and introduce new features. Test updates in a lab environment before deploying them to production systems.

Backup Power: maintain backup power solutions (UPS or generators) to ensure network continuity.

Monitoring and Alerts: Use Nagios network monitoring tools to track device health and performance. Set up alerts for hardware failures, unusual traffic. Perform proactive troubleshooting based on trends and logs.

Hardware Repairs and Replacements: Identify faulty components and replace them promptly to minimize downtime. Use spare parts or backup devices for critical networking hardware.

Warranty and Vendor Support: Keep track of warranties for all networking components. Maintain

contact with vendors or manufacturers for support and replacement of faulty hardware. Consider extended warranty or maintenance contracts for critical hardware.

Disaster Recovery and Spare Parts Management: Maintain an up-to-date disaster recovery plan to quickly restore network operations. Stock critical spare parts, such as power supplies, network cards, or backup hardware.

Scheduled Maintenance Plan: Daily: Monitor logs and alerts. Weekly: Check network connections and backup systems. Monthly: Update firmware and perform detailed hardware inspections.

Quarterly/Annually: Review the entire network setup and upgrade outdated hardware.

92) What exactly do you monitor on the device?

Ans ⇒ system resources, logs, security, availability, stability and performance metrics are the main areas of monitoring.

Hardware & System Resource

CPU & load Usage: top Memory Usage: free -m, top

Disk Usage & I/O: df -h, du -sh, iostat, lsblk Processes: ps, top, pidstat

Network

connectivity: - ping,

Network Traffic: ifconfig, ip add, traceroute.

Active Connections (List open ports and services): - netstat -tunlp, ss -tunlp Packet Analysis: tcpdump, traceroute.

Log Monitoring

System Logs: journalctl -xe, dmesg, /var/log/syslog, /var/log/messages,

Authentication Logs: /var/log/auth.log, /var/log/secure

Application Logs: /var/log/nginx/access.log, /var/log/mysql/error.log

Server & Application

Web Server Stats: - systemctl

Application performance: - latency, error logs, error messages, auto scaling. Container Monitoring:

docker - ps, docker -stat

Server: - Nagios, Prometheus, Grafana.

93) What exactly are you check & troubleshooting? What do you escalate?

Ans ⇒ I troubleshoot issues related to:

System Performance (CPU, RAM, Disk usage, latency) Network Connectivity (Slow speeds, packet loss, firewall issues)

Application Failures (Web servers, databases, docker microservices, latency) Security Incidents (Unauthorized access, changes in SG, failed logins) Service Availability (Down services)

Log Analysis (Error messages, kernel panics, access logs)

In the alert we check what is the CPU & load percentage utilized. In the alert we check what Disk and RAM memory are consumed and how much GB free storage is available. In the security Group we check which port number is added or deleted and by whom. User activity add and remove port, policy changes, add and remove user, console login failure alert. Check the latency, VPN down, Failed SSH Logins.

I troubleshoot the issue using logs, performance metrics and network tools. I attempt to resolve common issues (restarting web server, clearing disk space, killing unnecessary processes, restart server in case of server down). If the issue is not resolved by me. I escalate those metrics to the L-2 team with the detailed Error logs, Error messages, metrics, action taken by me, time duration and impact analysis.

Sometimes we need to communicate with stakeholders clearly about the business impact, estimate time for the resolution via bridge call, email or team chat. I inform them that our team is working on it continuously with high priority to get the recovery. I follow an escalation and structured troubleshooting approach to ensure a resolution is reached as quickly as possible.

94) What is the difference between 400 errors and 500 error?

Ans ⇒ 400-series errors (4xx) → Client-side errors (issues caused by the user's request). These occur when the request is invalid or cannot be fulfilled due to a mistake by the client/user.

A 400 error is a "Bad Request" HTTP status code. It means that the server cannot process the request because the client (invalid URL browser, script, or API call) sent something invalid.

A 401 Unauthorized error occurs when a client tries to access a resource but fails authentication. This means the server expects valid credentials (username/password, API token, etc.) but they were missing or incorrect.

A 403 Forbidden error means the server understands your request but refuses to fulfill it due to permission issues. The resource exists, but access is denied.

A 404 Not Found error means Resource missing the requested URL or file does not exist on the server. (404 not found / Http 404 / Error 404 / 404 page not found)

A 405 Method Not Allowed error occurs when a client requests a resource using an HTTP method that the server does not allow. Example: A form that allows only POST, but the client sends a GET request.

A 408 Request Timeout error is an HTTP status code that means Request took too long or the server did not receive a complete request from the client within the expected time.

500-series errors (5xx) → Server-side errors (issues caused by the server's failure to handle the request). This occurs when the server fails to process a valid request due to internal issues.

A 500 Internal Server Error is a generic HTTP status code indicating that something has gone wrong on the server, but the server cannot specify what the exact issue is. It means the request was valid, but the server failed to process it due to an internal problem.

A 501 Not Implemented error means that the server does not support the requested method. A 502 Bad Gateway error means a gateway or proxy received an invalid response from an upstream server. This usually happens in a multi-server setup where one server depends on another to process the request.

A 503 Service Unavailable error means that the server is temporarily unable to handle the request. may the following reasons:

Server Overload - The server is receiving too many requests and cannot process them all.

Maintenance Mode - The server is down for scheduled maintenance or updates. Temporary Issues - There may be temporary network congestion, resource exhaustion (CPU/memory), or dependency failures (database being down).

A 504 Gateway Timeout error occurs when a server takes a long time to respond.

Slow or unresponsive server - The main server is taking too long to respond. Network issues between servers - Connectivity problems between the proxy and the upstream server.

High server load - The backend server is overloaded and can't respond in time.

security rules - Blocking communication between the servers.

Insufficient Storage- The server has run out of space due to Full disk or Too many logs/files.

95) What do you mean by server-side error and client-side error?

Ans ⇒ A server-side error and a client-side error refer to different types of issues that can occur when a client/user interacts with a server.

Client-Side Error: - issue on the user's end (web browser, application or device). issues caused by incorrect requests, missing parameters or unsupported operations. Represented by HTTP status codes in the 4xx range (e.g., 400, 403, 404).

400 Bad Request: The client sent an invalid request.

401 Unauthorized: Authentication is incorrect.

403 Forbidden: The client is not allowed to access the requested resource.

404 Not Found: The requested resource does not exist on the server.

408 Request Timeout: - the request took too long to respond.

Server-Side Error: - issue due to misconfigurations, resource limits, or unexpected conditions. Represented by HTTP status codes in the 5xx range (e.g., 500, 502, 503).

500 Internal Server Error: A generic error when something goes wrong on the server.

502 Bad Gateway: The server received an invalid response from an upstream server.

503 Service Unavailable: The server is temporarily overloaded or undergoing maintenance.

504 Gateway Timeout: The server did not receive a response in time from another server.

Client-side errors (4xx) mean the issue is with the request from the client.
Server-side errors (5xx) mean the issue is at the server's end.

96) How & what do you verify whether it is a DB issue or third-party issue?

Ans ⇒

DB Connectivity:

connect to the DB using mysql, psql.

Make sure DB port open (netstat -tulnp | grep <DB_PORT>)

Verify application Connect to the DB (telnet <DB_HOST> <DB_PORT>), Check firewall rules.

DB Load and Performance: - Check DB CPU and memory usage (top, free -m).

Disk Space and I/O: - Ensure the DB has enough disk space (df -h and du -sh /var/lib/mysql or /var/lib/postgresql). Check disk I/O performance

Logs and Errors

MySQL: /var/log/mysql/error.log

PostgreSQL: /var/log/postgresql/postgresql.log

97) Do you have experience with the configuration level?

Ans ⇒ yes, ssh configure, user configure, host & service configure, Sudo user configures, Apache configure, permission management, disk management, logs management,

“ /usr/local/nagios/etc/objects/jenkins.cfg ” add these file paths in the main nagios.cfg file

“/usr/local/nagios/etc/object/jenkins.cfg v /usr/local/nagios/etc/nagios.cfg” compare file. Logs ⇒ /var/logs/nagios/access.logs etc.

98) The route call: how do you identify?

Ans ⇒ verify Network Routing Issue (Data Traffic): - If the issue is related to IP routing (users can't reach a destination, high latency & packet loss).

1: Connectivity: - Use ping to check if the destination is reachable or not. Success: The network route is likely fine.

Failure: There may be a route issue, firewall block, or ISP problem.

2: Network Route (Traceroute / MTR): - To check how traffic flows through the network. If packets stop at a specific hop, the issue is at that network point. If there is high latency at a specific hop, it could indicate congestion.

3: Routing Tables (Linux/Router)

Linux server: - ip route show

If the destination network is missing, the route is not set. If the wrong gateway is used, packets are misrouted. Fix:

Update the route using: - ip route add.

99) What is the emergency option from my programming language?

Ans ⇒ In a NOC emergency shutdown and recovery procedures are critical to prevent data loss, service downtime and infrastructure damage.

When to Perform an Emergency Shutdown?

A shutdown should only be done, if necessary, such as:

Critical Hardware Failure - Overheating, power supply failure, or disk corruption.

High CPU & Load Spikes - If the system is unresponsive and affecting critical services. Security Breach - In case of a cyberattack (unauthorized access).

Data Center Power Failure - When UPS/generators are at risk of failure. Hardware Maintenance - If an immediate shutdown is required to avoid damage

ServiceNow and Jira - Common in enterprise ITSM (IT Service Management).

100) Active directory, Patch, User, File and permission management in Linux?

Ans ⇒ **Active Directory** is a Microsoft service that manages users, devices, and network resources for Windows domain networks. AD verifies users' credentials and determines what files and applications they can access.

In the Linux the logs files are stored in the var/logs folder. File: /var/log/syslog or /var/log/messages.

pwd - command in Linux is that displays the full pathname of the current working directory. pwd or pwd -

L Shows the symbolic (logical) path. pwd -P Shows the physical path.

The cd command in Linux is used to change the current working directory. Absolute Path: Specifies the full path from the root's directory.

Relative Path: Specifies the path relative to the current directory.

cp command in Linux is used to copy files and directories from one location to another. cp resume.txt /root/home/kapil

-r or -R: Copy directories and their contents recursively.

cp -r /source_directory /destination_directory

The touch command in Linux is primarily used to create a multiple empty file or update the timestamp of an existing file. touch file.txt

-a: Change the access time only. touch -a file.txt

-m: Change the modification time only.

-c: if file does not exist for modification, avoid creation of a new file.

-d: Set the custom time and date; touch -d "2024-04-21 13:40" file.txt

-t: Set the custom time.

rm command used to remove files and directories. Remove the file if you have permission: rm file.txt

-f: forcefully when you don't have permission. rm -f file.txt

-r: mandatory directory with subdirectories. rm -rf /kp or rmdir /kp

-i: ask confirmation before each file or directory is removed

-d: preventing the accidental deletion of non-empty directories.

Dig (domain information groper) -It provides detailed information about domain names, IP addresses & DNS. It is commonly used for troubleshooting DNS issues and obtaining detailed DNS information.

-t Specifies the type of DNS record to query. dig -t CNAME example.com

DNS-

A ⇒ ip4 address,

AAAA ⇒ ip6,

CNAME⇒ alias ip4/6 name to main-domain-name

User Management

create a new user directly in dev group: - sudo useradd -mG groupname -s /bin/bash kapil create & delete a user account - useradd kapil, userdel kapil.

-r: Deletes the user's home directory and mail spool.

create & delete a group - groupadd devops, groupdel devops

usermod: - modify an existing user account: usermod -aG groupname username

-g: Changes the user's primary group.

-G: Changes the user's secondary groups.

-a Add to supplementary groups without removing them from other groups.

modify existing group names. groupmod -n new old-name

gpasswd - Manages group password, assigns admin add/del user from group.

-a Add a user to a group. gpasswd -a user group

-R restricts access to the group.

-d Remove a user from a group.

-r Remove the group password.

Groups: Show Group Membership: group Kapildeo

Chgrp - change ownership of a group. chgrp root /file or folder. Chown - change ownership of a user.

chown root /file or folder. Chmod - change permission of a file and directory. chmod 777 file.txt

-R Change ownership recursively for all files & directories within a directory.

passwd - update authentication

-d : Deletes a user's password.

-e : Force users to change their password the next time they log in.

-l : Locks the user's password.

-u : Unlocks the user's password.

-S : Displays the account's password status.

User default setting /etc/default/useradd GROUP=100

HOME=/home INACTIVE=-1 EXPIRE=

SHELL=/bin/bash SKEL=/etc/skel CREATE_MAIL_SPOOL=yes

Password age rules /etc/login.defs PASS_MAX_DAYS 90 # Max days a password is valid
PASS_MIN_DAYS 7 # Minimum days between password changes
PASS_WARN_AGE 14 # Days before expiration to warn user
chage is used to change the user password expiry info. chage -d -m 10 -M 90 -W 10 -I 3 -E 1 kapildeo
-d: Sets the last day the password was changed.
-E: Set the account expiration date: sudo chage -E 2024-12-31 john
-I: Sets the number of inactive days before the account is locked.
-m: Sets the minimum number of days required to change password.
-M: Sets the maximum number of days between password changes.
-W: Sets the number of days of warning before the password expires.
The whois command in Linux is used to query information about a domain name & IP address. It contacts a WHOIS server to retrieve details like domain registration data, ownership information, registrar, and more.

101) How can we check open ports in Linux?

Ans ⇒ netstat -tuln: it is used to monitor incoming and outgoing network packets. It displays network connections, routing tables, interface statistics and listening ports. We can check custom port number of ssh (netstat -tulnp | grep 22)
-p: display PID & program name, helps us to see which process is using a specific port.
-t: list TCP connections.
-u: list UDP connections.
-l: ports that are waiting for incoming connections.
-r: Display routing table.
-i: Display interface statistics.
ss (Socket Status): Display detailed information about address, socket & port connections. ss -tuln : display listening TCP and UDP sockets.
ss -a: Show all open sockets
lsof: Shows which files are opened by which process. lsof -u username
lsof -p <PID>

102) How to kill the process forcefully?

Ans ⇒ The kill command sends a signal to a process to terminate it.
For example, kill -9 pid forcefully stops a process. pkill stop with process name.

103) What is a shell script?

Ans ⇒ A shell script is a file containing a series of commands that can be executed by the shell. It's used for automating tasks.

104) How do you schedule a cron job?

Ans ⇒ The crontab command in Linux is used to manage cron jobs, which are scheduled tasks that run at specific time intervals. crontab -e used to edit the cron table and schedule tasks.
-e: Edit the current user's crontab file.
-l: List the current user's crontab file.
-r: Remove the current user's crontab file.
-i: confirmation before removing the crontab
-u: Set a crontab file for a different user.
This option is typically used by the root user ⇒ crontab -u username -e
Nagios log ⇒ /usr/local/nagios/var/nagios.log ⇒ /usr/local/nagios/var/archive/nagios.26-08-2024.log
Crontab -e ⇒ 0 6 * * * mv /usr/local/nagios/var/nagios.log /usr/local/nagios/var/archive/ format: minute-hour-date-month-weekday ⇒ 30 2 * * * /home/user/scripts/backup.sh

105) What is LVM in Linux?

Ans ⇒ LVM is a method of allocating space on storage devices that is more flexible than conventional partitioning schemes.
Logical Volume Manager: - It is a flexible disk management system in Linux that allows you to create, resize, and manage disk partitions dynamically – even while the system is running. Check package: - rpmquery lvm2

106) What is the purpose of Sudo in Linux?

Ans ⇒ Sudo allows a permitted user to execute a command as a superuser or another user, as specified by the security policy.

107) What is the /etc/fstab file?

Ans ⇒ This file contains information about disk drives and partitions, and it is used by the system to automatically mount file systems at boot.

108) How do you check your IP address?

Ans ⇒ Ifconfig: - Used to configure and display network interface parameters. It displays the configuration of all network interfaces, IP address, netmask, broadcast address, and interface status. Ifconfig eth0 - This shows details of eth0 interface.

Assign IP to eth0: sudo Ifconfig eth0 192.168.1.100 netmask 255.255.255.0 Enable or disable an interface: sudo Ifconfig eth0 up & sudo Ifconfig eth0 down

Ip address: A more modern command to display and manage network interfaces, routing etc. Display all IP addresses: ip addr

Display a specific interface: ip addr eth0

Assign IP to eth0: Sudo Ip address add 192.168.1.100/24 dev eth0

Bring an interface up or down: Sudo ip link set eth0 up & Sudo Ip link set eth0 down

nslookup - The nslookup command is a network administration tool used to query DNS (Domain Name System) servers for information about domain names and IP addresses. It provides a way to obtain DNS records and troubleshoot DNS-related issues.

Lookup a domain name: nslookup google.com Lookup an IP address: nslookup 198.168.0.200

109) How do you view running processes in Linux?

Ans ⇒ Top - The top command in Linux provides a real-time, dynamic view of the system's resource usage, including CPU, system Load, memory, and process information. It's a powerful tool for monitoring system performance and identifying resource hogs.

I ⇒ display running process & hide sleeping process.

I ⇒ display the number of core CPUs. (cat /proc/cpuinfo). Shift + P ⇒ to sort processes by CPU usage.

Shift + M ⇒ to sort processes by Memory usage.

nice ⇒ it is used to set the priority of a process before it starts. It allows you to control the "niceness" of a process. The niceness value ranges from - 20 (highest priority) to +19 (lowest priority). A lower niceness value gives the process more CPU time, while a higher value gives it less. nice -n 15 command (tar -czf archive.tar.gz /path/to/directory)

ctrl + z ⇒ stop background process.

Jobs ⇒ check background status running or stopped.

bg ⇒ After suspending a job with ctrl + z , bg used to continue to run in the background.

System Summary:

Uptime: How long has the system been running. Users: Number of logged-in users.

Load Average: System load averages for the past 1, 5 and 15 minutes. Tasks: Number of tasks (processes) running, sleeping or stopped.

CPU: %CPU usage by user processes, system processes, and idle time. Memory Usage: Total, used, and free memory and swap space.

Process List:

PID: Process ID, name.

USER: User who owns the process. PR: Process priority.

NI: Nice value (priority adjustment).

VIRT: Virtual memory used by the process

RES: Resident memory (physical memory) used by the process. SHR: Shared memory used by the process.

S: Process state (sleeping, running, inactive).

%CPU: CPU usage percentage by the process.

%MEM: Memory usage percentage by the process. TIME+: Total CPU time used by the process.

COMMAND: Command or process name.

h: Display help screen q: Quit top.

P: Sort processes by CPU usage.

M: Sort processes by memory usage. T: Sort processes by time.

k: Kill a process (prompt for PID and signal). kill -9 pid ⇒ forcefully r: Renice a process (change its priority).

u: sort processes by user.

Top shows the percentage of CPU time spent on various tasks: -

us (User Space): ⇒ This indicates that 12.0% of the CPU time is being used by user processes, which are non-kernel processes initiated by users or applications.

sy (System Space): ⇒ This shows that 3.0% of the CPU time is being used by system processes, which are the kernel processes handling system tasks.

ni (Nice Value): ⇒ This represents the percentage of CPU time spent on processes with a manually adjusted "nice" value, which affects their priority.

id (Idle Time): ⇒ This shows that 85.0% of the CPU time is idle, meaning the CPU is not doing any work during this time.

wa (I/O Wait): ⇒ This indicates the percentage of CPU time spent waiting for I/O operations (like disk reads/writes) to complete.

hi (Hardware Interrupts): ⇒ This shows the percentage of CPU time spent handling hardware interrupts, which are signals sent by hardware devices requiring the CPU's attention.

si (Software Interrupts): ⇒ This represents the percentage of CPU time spent handling software interrupts, which are similar to hardware interrupts but generated by software.

Zombie - The process has finished execution, but its record still lingers in the process table as a "zombie" process.

ps command ⇒ show information about all processes running on the system: ps -e or ps -A. For a specific user: ps -u username.

specified PID: ps -p 1234 ps -eo pid, comm, nice ⇒ check the pid, command, nice value.

lsof - monitors system input/output devices and provides statistics on CPU utilization and I/O activities for devices, partitions, and network filesystems. lsof is particularly useful for identifying performance related to disk and cpu.

-c = cpu, -d = device, -m = megabyte, -p = specific device

dmesg - used to display the kernel ring buffer messages, which often include hardware-related logs. So, we can find hardware related errors.

filter out hardware errors you can use - dmesg | grep -i error

we can check for specific terms like "fail" or "fault": dmesg | grep -iE 'error | fail | fault'

-i ⇒ ignore case : -E ⇒ extended regular expressions.

fdisk- it is used in Linux to manage disk partitions. It allows users to create, delete, modify and view the partitions on a hard drive. It's a powerful tool for disk management, particularly for creating or adjusting the layout of partitions before installing an operating system or for managing additional storage devices.

need to specify the disk name. sudo fdisk /dev/sda

Press n to create a new partition

Press d to delete a partition and then specify the partition number.

After creating or deleting partitions,

press w in the interface to write the changes.

Press q to quit without saving changes.

blkid ⇒ uuid ; lsblk ⇒ list hard drive.

scp (secure copy): - is used to securely transfer files and directories between hosts on a network. It uses SSH to ensure that the data is encrypted during the transfer.

Local to Remote ⇒ scp localfile.txt username@remotehost:/remote-destination-path

Remote to Local ⇒ scp username@remotehost:/remote-file-path /local-destination-path -p 100 specify the port number, if you set custom port 100 for ssh.

TFTP (port 69) is a simple file transfer protocol used primarily for transferring small files, such as boot files or configuration files, between devices on a network. Unlike FTP, TFTP does not provide authentication or encryption, making it less secure but more lightweight.

HTTPS (443) => This port number is used to facilitate encrypted communication between a web browser and a server

HTTP => 80 => Port 8080 is a port number used for web servers, and is often used as an alternative to port 80 for HTTP services.

FTP => 20 FTP port for data transfer between client and server. 21 FTP port for establishing connection between two hosts. It's referred to as the control port.

SMTP => 25 (Simple Mail Transfer Protocol). A protocol used by mail servers to send and receive mail.

DNS => 53 (Domain Name System) Responsible for resolving a domain name to machine-readable IP addresses or translating domain names into IP addresses.

110) How can we see the content of the file?

Ans => There are multiple commands to view the content of a file: cat, more, less, page, grep, tail, head, vi, vim and nano. Cat command is famous for viewing the content of a file: cat file.txt multiple files into one: cat file1.txt file2.txt > combined.txt Create new file: cat > file.txt

111) How do we manage the services in Linux?

Ans => for managing the services there is a Systemctl command. By using this command we can start, stop, restart, enable and disable the services as per requirement.

112) Server is rebooting continuously how will you check the reason for the last reboot?

Ans => Last.

113) What is RAID?

Ans => RAID (Redundant Array of Independent Disks) is a data storage virtualization technology that combines multiple physical disk drive components into one or more logical units. The goal is to improve performance & increase data redundancy.

RAID 0 (Striping) => Purpose: Performance

Data is split across multiple disks in chunks (stripes), which improves r/w operation. Advantages: Increased performance due to parallel read/write operations.

Disadvantages: No redundancy; if one disk fails, all data is lost.

RAID 1 (Mirroring) => Purpose: Redundancy

Data is duplicated on two or more disks. Each disk contains an identical copy of the data. Advantages: High redundancy and fault tolerance; if one disk fails, data is still available. Disadvantage: Increased cost due to having multiple copies of the same data.

RAID 10 (1+0) => Purpose: Performance and redundancy

Combine RAID 1+ 0. Data is mirrored (RAID 1) and then striped (RAID 0) across multiple disks.

Advantages: Offers the redundancy of RAID 1 and the performance of RAID 0.

Disadvantages: More expensive due to the need for twice as many disks. Key Concepts Striping:

Splitting data into blocks and writing them across multiple disks.

Mirroring: Duplicating data on multiple disks for redundancy.

Parity: Error-checking data used to reconstruct lost data in the event of a disk failure. Fault Tolerance:

The ability to continue operating in the event of one or more disk failures.

114) Difference between switch and router?

Ans => switch: - connects devices within the same network, operating on layer 2.

Router: - connects different networks, route data between them, operates at layer 3.

115) What are router and switch?

Ans ⇒ A router is a device that connects different networks and forwards data based on IP addresses. It is used for Internet access and network communication between different subnets. Example: A home router connects your Wi-Fi network (LAN) to the Internet (WAN).

Commands: - Cisco Router: show ip route

Linux Router: ip route show

A switch is a device that connects multiple devices within the same network (LAN) and forwards data using MAC addresses.

Exa: switch connects multiple computers, printers, and servers within the same office network.

Commands to Check Switch Configuration

Cisco Switch (Show MAC Table): show mac address-table

Use a Router to connect different networks (e.g., Internet, branch offices).

Use a Switch to connect multiple devices within a LAN (e.g., office computers).

Router: Think of it as a traffic controller that directs network traffic.

Route Table: Acts like a map the router uses to find the best path.

116) What is the OSI model? Explain each layer?

Ans ⇒ Layer Name Function Protocols / Devices

7 Application Provides user interface & network services. HTTP/S, FTP, SMTP, DNS

6 Presentation Translates, encrypts, and compresses data. SSL, TLS, JPEG, MP4

5 Session Establishes, manages, and terminates sessions.

4 Transport Ensures end-to-end communication and reliability. TCP, UDP

3 Network Finds the best path for data transfer (Routing). IP, ICMP, Routers

2 Data Link Handles error detection and MAC address. Ethernet, Switches, Bridges

1 Physical Transmits raw bits over a physical medium. Cables, Hubs, Wi-Fi, Fiber.

Physical Layer-1 Function: Transmits raw data through cables or wireless signals. Devices: Ethernet cables, Hubs, Fiber optics, Wi-Fi signals.

Example: Sending binary signals (0s and 1s) through a network cable.

Data Link Layer-2 Function: Provides MAC addressing and error detection for frames. Divisions: LLC (Logical Link Control): Handles error control.

MAC (Media Access Control): Assigns MAC addresses. Devices: Switches, Bridges, NIC (Network Interface Card).

Example: A switch uses MAC addresses to forward packets inside a LAN.

Network Layer-3 Function: Performs IP addressing and routing between networks. Uses IP addresses to forward packets.

Routing protocols (OSPF, BGP) determine the best path. Devices: Routers, Switches.

Example: A router forwards data between 192.168.1.1 (LAN) and 8.8.8.8 (Google DNS).

Transport Layer-4 Function: Ensures end-to-end communication and reliability. Protocols: TCP: Reliable, connection oriented.

UDP: Fast, connectionless.

Example: TCP ensures data integrity in an email, while UDP streams video with low latency.

Session Layer-5 Function: Manages, establishes, and terminates sessions between applications.

Example: When logging into a remote desktop session (RDP), this layer maintains the connection.

Presentation Layer-6 Function: Handles data formatting, encryption, and compression. Encryption: SSL/TLS secures HTTPS websites.

Compression: JPEG, MP4, GIF formats reduce file sizes.

Application Layer-7 Function: Provides services for user applications. Protocols: HTTP, HTTPS, FTP, SMTP, DNS.

Web browsing: Uses HTTP/HTTPS.

Email communication: Uses SMTP, IMAP, POP3.

117) What is SELinux?

Ans → SELinux (Security-Enhanced Linux) is a security architecture integrated into the Linux kernel that provides mandatory access control (MAC). It helps enforce security policies that restrict what users and programs can access and modify, even if they have root privileges.

To restrict programs and users to only the resources they absolutely need – even if they are root. This limits damage from compromised or misbehaving software.

In Enforcing mode: The access is blocked.

In Permissive mode: The access is allowed but logged.

Enforcing: - SELinux policy is enforced. (Access denied if rules don't allow it): SELINUX=enforcing

Permissive: - Policy violations are logged but not enforced. Useful for debugging: SELINUX=permissive

Disabled: - SELinux is turned off. No policy enforcement or logging: SELINUX=disable

setenforce 0 # Switch to Permissive setenforce 1 # Switch back to Enforcing

118) What is kernel?

Ans ⇒ The kernel is the core component of an operating system. It acts as a bridge between hardware and software, managing system resources and enabling applications to interact with hardware safely and efficiently.

119) How to copy data from one server to another?

Ans ⇒ cp for local, one path to another path scp for one server to another, copy data which was already copied previously. rsync one server to another, only copy the file if the data updates or create a new file. It uses ssh protocol for transferring data.

120) What is an ip-table & firewall?

Ans ⇒ firewall - Modern Firewall Manager, iptables - Traditional Firewall Tool

Packet filtering (allow or block traffic) Network address translation (NAT) Port forwarding Securing servers

List all rules in the filter table sudo iptables -L -v -n # Allow incoming SSH connections

sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT # Block all incoming traffic by default sudo iptables -P INPUT DROP

Delete a rule

sudo iptables -D INPUT -p tcp --dport 22 -j ACCEPT

To block all incoming connections except SSH and HTTP:

sudo iptables -P INPUT DROP

sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

sudo iptables -A INPUT -m state --state ESTABLISHED, RELATED -j ACCEPT

121) How will you resolve the web server issue, If the URL is not responding?

Ans ⇒ Verify Server is Running: Ensure the server is powered on and the web server software Apache, Nginx service is running. Tools like systemctl.

Monitor Resource Usage: Check the server's CPU, RAM, and disk space usage. High resource consumption can cause the server to become unresponsive. Tools like top.

Review Server Logs: Server logs are a treasure trove of information when troubleshooting. Access Logs: These logs record all requests made to the server. They can help identify if requests are reaching the server at all.

Error Logs: These logs detail errors encountered by the web server software. Look for any recent entries that coincide with the time the URL became unresponsive. The location of these logs varies depending on the web server and operating system (/var/log/apache2/error.log for Apache on Ubuntu).

Network connectivity: -

Ping: - ping command followed by your domain name to check if the server is reachable on the network.

Traceroute: - check network path from client to the server, helping to identify any points of failure along the way.

DNS Resolution: Use tools like nslookup or dig to verify that the domain name is resolving to the correct IP address. For example, nslookup example.com. If the IP address is incorrect, you'll need to update your DNS records at your domain registrar or DNS hosting provider.

Firewall: - Ensure that firewalls on the server or within your network are not blocking the standard web traffic ports (Port 80 for HTTP and Port 443 for HTTPS).

122) What is Server Hardening?

Ans ⇒ Server hardening is the process of securing a server by reducing its attack surface. The primary goal is to minimize security vulnerabilities. Configuring systems securely Applying updates and patches Enforcing access controls. Remove unused software, packages and services. Disable unused user accounts. Keep the OS and kernel updated with patches. Set up automatic security updates (if suitable).

User Account and Authentication

Disable root login over SSH (PermitRootLogin no).

Use SSH key-based authentication instead of passwords. Enforce strong password policies. Use tools like sudo to limit root access.

Network Hardening: - Block unused ports using a firewall. Enable only essential services (HTTP, HTTPS, SSH).

File System and Services

Set correct file permissions and ownership. Security Tools & Logging

Install SELinux for mandatory access control, FirewallD to control access

123) What step will you take if the server goes down?

Ans ⇒ I will check alerts & logs then I will send that alert on to the team's channel. I will ping the server to check the connectivity I will try remote access (ssh). Check power and hardware status and Restart services or servers if needed. Escalate if the issue persists.

124) What is traceroute & how is it used?

Ans ⇒ traceroute is a network diagnostic tool used to track the path packets take from source to the destination. It helps to identify where the delay and failure occurred in the network.

How to reduce storage size by using the LVM, XFS file system used on that partition? Ans ⇒

As mentioned, XFS cannot be shrunk. The only procedure is to recreate it.

If you're using XFS and still want to reduce the LVM volume size, you'll need to:

Backup your data.

Delete and recreate the logical volume (LV) with a smaller size.

Restore your data.

125) How to run the sh file?

Ans ⇒ if the file present in the current location = ./kp.sh

if the file present in the different location = sh /home/ec2user/kapil/kp.sh

if the file present in the different location and you are in the path just provide = sh kapil/kp.sh

126) how to check the current working shell in Linux?

Ans ⇒ default shell cat /etc/passwd (alice:x:1001:1001::/home/alice:/bin/bash) Current shell echo \$SHELL (/bin/bash) it will print the current shell.

Current shell ps -p \$(pgrep -u username -n) -o comm= it will print the current shell.

127) How to list hidden files?

Ans ⇒ l. Command to list hidden files only. ls -a Command to list hidden & other files. ls -ltr list all with detailed information.

128) What Provides Centralized Authentication in Linux?

Ans ⇒ 1. LDAP (OpenLDAP) Lightweight Directory Access Protocol

Stores users, groups, passwords in a directory tree. Used for authentication across multiple systems.

129) How to change the umask value?

Ans ⇒ add umask value in the .bashrc file (umask 000) for the user. For the global vi /etc/bashrc (mention umask 000 or umask -S o=rwx, g=r, o=r)

130) What is the use of ACL?

Ans ⇒ assign special permission for the individual user or individual group.

131) What is a dry run?

Ans ⇒ A dry run is a simulation of a command or process – it shows what would happen without making any changes.

It's a "safe test"

No files are changed; no configurations are applied – it's just a preview.

rsync --dry-run: -Shows what files would be copied or deleted.

yum, dnf --assumeno: - Shows what would be installed/removed.

Avoids unintended changes or data loss.

Help you validate your commands before applying them. Essential for production environments and automation scripts.

How to check when the rpm package is installed? Ans ⇒ rpm -qi <package-name>

rpm -qi http

Q1: Your server has suddenly become unresponsive. Users are complaining they can't access services. What steps would you take to troubleshoot and bring the server back to a healthy state?

Answer: First, I'd check if the server is reachable using ping, or telnet <host> <port> for specific service checks. If remote, I'd confirm VPN or SSH access is working.

Next, I'd check server health: uptime, top, free -m, and df -h to verify CPU, memory and disk. systemctl status nginx or apache2 to see if the web server is active.

Then, I'd ensure the correct ports are open in the OS firewall (firewalld, iptables) and in the cloud security group/inbound rules.

Finally, I'd check logs like /var/log/messages, /var/log/nginx/error.log, or app-specific logs to look for errors. If everything infra-side looks clean, I'll coordinate with the developer team.

Q2: How do you manage log rotation in Linux? What tool is used? How do you configure it to rotate daily at 11:59 PM?

Answer: - To manage log rotation in Linux, we use the logrotate utility. For service-specific rotation, I create a config in /etc/logrotate.d/<service> and define parameters like daily, rotate 7, compress, etc.

If I need it to run at 11:59 PM, I will create a custom cron job:

```
59 23 * * * /usr/sbin/logrotate /etc/logrotate.conf
```

For centralized logging, I configure /etc/rsyslog.conf or a file in /etc/rsyslog.d/ with:

```
*.* @<central_log_server_ip>:514
```

On the server side, I ensure port 514 is open and configure the rsyslog to accept incoming logs by uncommenting:

```
module(load="imtcp") input (type="imtcp" port="514")
```

Then I restart the rsyslog service on both ends:

```
systemctl restart rsyslog
```

Q3: Your disk is filling up. How would you find which files or directories are using the most space, and what actions would you take?

Answer: First, I'll check overall disk usage with: df -h

Then, I'll find which directories are consuming the most space: du -sh /* 2>/dev/null du -ah / | sort -rh | head -20

Once I identify the space-consuming data, I'll coordinate with the respective user or owner to check if it's needed. If it's not important, I'll safely delete it. If it's important, I'll consider increasing the volume (via AWS EBS or LVM) temporarily. Before any cleanup, I'll ensure a proper

backup is taken to avoid data loss.

Q4: A user reports that they can't log in to the Linux server via SSH. How would you troubleshoot and resolve this issue?

Answer: First, I'll ensure the user is connected via VPN if the server is on a private network, and that their internet connection is stable.

Next, I'll check: Whether port 22 (or custom SSH port) is open in the AWS security group/inbound rule. Whether firewall/iptables is allowing the port on the server, I'll check if SSH is running: `systemctl status sshd`

`ss -tulnp | grep ssh`

Then I'll open the SSH config file `/etc/ssh/sshd_config` and check: If `PasswordAuthentication` is set to yes (for password auth)

Or if `PubkeyAuthentication` is set to yes (for key based) If needed, I'll check the logs:

`tail -f /var/log/secure # RHEL tail -f /var/log/auth.log # Ubuntu`

Lastly, I'll verify the user account:

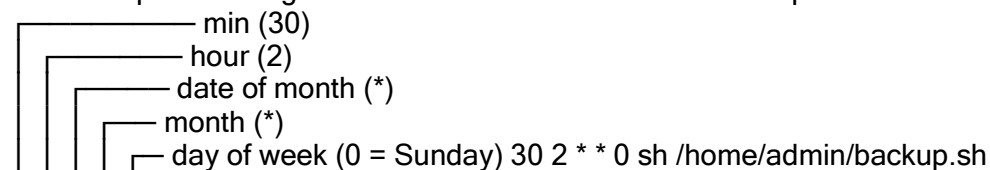
Is not locked or expired (`passwd -S username, chage -l username`)

Has a valid shell.

Q5: How would you schedule a cron job that runs a script `/home/admin/backup.sh` every Sunday at 2:30 AM?

Ans ⇒ Cron Syntax: `- 30 2 * * 0 /home/admin/backup.sh`

Or to be specific using sh: `- 30 2 * * 0 sh /home/admin/backup.sh`


`30 2 * * 0 /bin/bash /home/admin/backup.sh`

Make sure your script is Executable: `chmod +x /home/admin/backup.sh` Uses full paths inside the script.

Question 6: You've just logged into a critical production server, and you notice that the load average is unusually high. Can you explain step by step to investigate and identify the root cause of this issue?

Answer: When I see a high load average, I follow a systematic approach to narrow down the cause.

First, I'd confirm the average load using the `uptime` command. This also tells me how long the server has been running. A high load on a server that just rebooted is different from one that's been stable for months.

Next, I'd use `top` or `htop` as my main dashboard. This gives me a real-time view of what's happening. I immediately look at the average load values themselves (for 1, 5, and 15 minutes) to see if the problem is recent or ongoing.

The CPU stats line, specifically the `%wa` value. If I/O wait (`%wa`) is high, it points towards a disk or network I/O problem, not a CPU-bound one.

The list of processes sorted by CPU (`%CPU`) or Memory (`%MEM`) to see if a specific process is a resource hog.

If the CPU is high (`%us` or `%sy`): I'd identify the top process. Using its PID, I might use a tool like `strace -p <PID>` to see what system calls it's making, which can reveal what it's doing. I would also check the application's own logs.

If I/O Wait is high (`%wa`): This tells me the CPU is waiting for data from the disk. I would then use tools like `iostat -xz 1` or `iotop` to see which disk is being heavily used and which process is responsible for the heavy reads/writes.

If Memory is the issue: If I see high swap usage on the output of `free -h`, it means the system is out of RAM. I'd use `top` sorted by memory to find the memory-leaking application.

Finally, I'd check the logs. I would look at `/var/log/messages` or use `journalctl` to check for any kernel errors, hardware issues, or other system-level problems that might be contributing to the load.

Question 7: A junior administrator on your team has created a shared directory for a project where multiple users need to add and modify their own files. However, they are running into an issue where

users are accidentally deleting files that belong to other users within that same directory.

Answer: -"The correct solution is to set the sticky bit on the shared directory.

What it Does: When the sticky bit is set on a directory, it restricts file deletion. A user can only delete or rename a file within that directory if they are the owner of the file, the owner of the directory, or the root user. This is perfect for a shared folder, as it allows everyone to create files but prevents them from deleting files owned by others.

How to Set It: I would use the `chmod` command. There are two common ways to do this: Symbolic Method: `chmod +t /path/to/shared_directory`

Octal Method: The sticky bit is represented by the number 1 in the first (special permissions) octal position. So, if the directory permissions were 777 (read, write, execute for all), I would set them to 1777. The command would be `chmod 1777 /path/to/shared_directory`.

Verification: After setting it, I would verify the permissions using `ls -ld`. The output would look something like `drwxrwxrwt`, where the `t` at the end signifies the sticky bit is set.

Question 8: You are tasked with securing a new web server. You need to list all the services that are currently listed on TCP ports to ensure there are no unauthorized open ports.

First, what `netstat` command would you use to see all listening TCP sockets, their port numbers, and the names of the processes using them?

Second, is there a more modern command you might prefer to use instead of `netstat`, and what are its advantages?

Answer: - "For the first part of the question, using the classic `netstat` command, the command I would use is `netstat -tulnp`.

-t: Show TCP connections.

-u: Show UDP connections.

-l: Show only listening sockets.

-n: Show numeric addresses and port numbers, which is faster than trying to resolve hostnames.

-p: Show the process ID (PID) and name of the program that owns the socket.

For the second part, the more modern and preferred command is `ss` (socket statistics). The `ss` command has several key advantages over `netstat`:

It is much faster. `ss` gets its information directly from kernel space via the `tcp_diag` module. `netstat` works by reading and parsing the `/proc/net/tcp` file, which can be very slow and resource-intensive on a server with thousands of active connections.

It provides more detailed information about the state of the connections.

The `ss` command to get the same information would be `ss -tulnp`. As you can see, the flags are conveniently very similar, making it an easy tool to switch to."

Question 9: You are working on a server that uses `systemd` as its init system. A critical service, for example, the `nginx` web server, is failing to start. You run `systemctl start nginx` and the command returns an error almost immediately.

What specific, `systemd`-native command would you use to view the logs and diagnose the reason for this startup failure? And how would you use it to see only the logs related to that specific service?

Answer: - "For a `systemd`-managed service, the primary, native command to investigate its logs is `journalctl`. It's the standard way to find out why a service isn't starting.

To see the logs for the specific `nginx` service, I would use the `--unit` or `-u` flag. The command is: `journalctl -u nginx.service`

To make troubleshooting easier, I would add a couple of other flags:

I'd use the `-e` flag to immediately jump to the end of the log output, so I can see the most recent error messages first. The command would be `journalctl -u nginx.service -e`.

If I wanted to watch the logs in real-time while I attempted to start the service again, I'd use the `-f` flag to follow the log. The command would be `journalctl -u nginx.service -f`.

Using `journalctl -u nginx` is the most direct way to see exactly what `systemd` has logged about the `Nginx` startup process, including any configuration errors or permission issues it encountered."